



DIVINE EDUCATION CENTRE
PRE-REGISTRATION EXAMINATION
2026

MATHEMATICS SET 3

Time allowed: 2 hours 30 minutes

Random No.						Personal No.		

Candidate's Name:

Candidate's Signature:

District ID No.

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Read the following instructions carefully:

1. Do not write your **school** or **district name** anywhere on this paper.
2. This paper has two sections: **A** and **B**. Section **A** has **20 questions** and **section B** has **12 questions**. This paper has **15 pages** printed altogether.
3. Answer **all** questions. All the working for both sections **A** and **B** must be shown in the spaces provided.
4. **All** working **must** be done using a **blue** or **black** ball point pen or ink. Any work done in pencil other than graphs and diagrams will **not** be marked.
5. **No calculators** are allowed in the examination room.
6. Unnecessary **changes** in your work and handwriting that cannot be easily read may lead to **loss of marks**.
7. Do not fill anything in the table indicated **“For**
Examiners' use only” and the boxes inside the question paper.

FOR EXAMINERS' USE ONLY		
Qn. No.	MARKS	EXR'S No.
1- 5		
6 -10		
11- 15		
16 – 20		
21 – 22		
23 – 24		
25 – 26		
27 – 28		
29 – 30		
31 – 32		
TOTAL		

SECTION A (40 MARKS)

Answer **all** questions in this section.

Question **1** to **20** carry two marks each

1. Work out; $111 \div 3$

2. Write 19,046 in words.

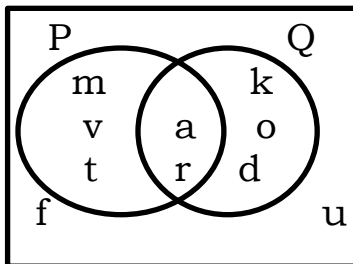
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3. Simplify; $2w - 3m + 5w - m$

4. Find the next number in the sequence;

-1, 2, -4, 8, -16, ____

5. Study the Venn diagram below carefully and use it to answer the question that follows.



Find $n(P \cup Q)^c$

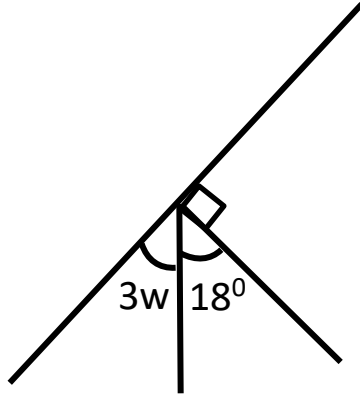




6. Given that  represents 120 mango trees planted by every home in a village. If the village has 18,000 mango trees altogether, how many homes are in that village?
7. Write 0.75 as a fraction in its lowest term.
8. Round off 89.86 to the nearest whole number.
9. With the help of a protractor, construct an angle of 55° in the space provided below.
10. Annet planned to go for a Church service which starts at 9:10 am. The journey from her home to Church takes her 20 minutes while walking. At what time should she leave home if she wants to reach Church five minutes before the service starts?



11. Find the value of w in the figure below in degrees.



12. John saves Sh. 3000 every day, how much money will he save in the whole month of February of a leap year?

13. Solve the equation;
 $m + 11 = 3m - 3$

14. Workout;
 $17 - 3^2 \times 2 + \sqrt{25}$

15. A shopkeeper sells cooking oil in 0.5 litre containers. How many such containers will he sell from a twenty litre jerrycan?



16. On a certain day, $\frac{1}{4}$ of the pupils were absent. The teacher served 3 sweets to each pupil present on that day. If the total number of pupils in that class was 40, how many sweets did the teacher serve?
17. The probability that Uganda Cranes will qualify for African Cup of Nations is 0.2. Find the probability that Uganda Cranes will not qualify for African Cup of Nations.
18. On the ground of a mountain, the temperature is 7°C and -2°C on top. Find the difference in temperature between the ground and top of the mountain.
19. The price of a watermelon increased by $15\frac{1}{2}\%$ to sh. 4620. What was the original cost of the watermelon before the increase?
20. A square tile covers a surface area of 0.25 square metres. Find the length of one side of the square tile.



SECTION B: 60 MARKS

Answer **all** questions in this section
Marks for each question are indicated in brackets

21. a) Work out:

$$\begin{array}{r} 312_{\text{five}} \\ \times 2 \\ \hline \end{array}$$

(02marks)

b) Given that $12_p = 8_{\text{ten}}$.
Find the value of p .

(02 marks)

c) Convert 412_{five} to a common base.

(02 marks)

22. The sum of 3 consecutive natural numbers is 24.
If the middle number is $y + 1$, find all the numbers.

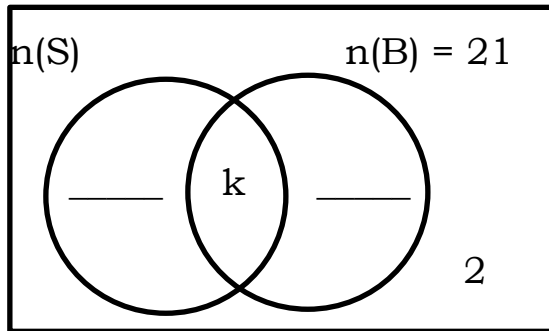
(04 marks)



23. At a party, 14 guests took soda only (S), 21 guests took beer (B), k guests took both Soda and beer while 2 guests did not take any of the two drinks.

a) Use the above information to complete the Venn diagram below.

(02 marks)



b) If 17 guests took Soda, find the value of k .

(02 marks)

c) How many guests attended the party?

(01 mark)

24. Mr. Kivumbi had his garden measuring 15 metres long and 10 metres wide. He hired someone to plough his garden at a cost of sh. 200 per square metre. How much did it cost him to plough the whole garden?
(04 marks)



25. At a certain Bank, the exchange rate for buying and selling foreign currencies were as shown below:

Currencies	Buying in Ugsh.	Selling in Ugsh.
1 United States dollar (USD)	3750	3800
1 Euro (€)	4600	4700
1 Kenya shillings (Ksh)	30	32

- a) If Mary had 400 United States dollars and exchanged for Uganda Shillings, how much in Uganda Shillings did she get? **(02 marks)**

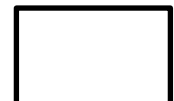
- b) John had €160 and wanted to buy a smart phone in Kenya Shillings. What was the cost of the phone in Kenya Shillings? **(03 marks)**

26. Two farmers harvested rice:
- Farmer A harvested 1,450 kg
 - Farmer B harvested 975 kg

They combined their rice and packed it equally into 25 bags.

a) How many kilograms were packed in each bag? **(03 marks)**

b) If they sold all the rice at sh. 3000 per kg, how much was given to farmer B according to his contribution? **(02 marks)**



27. a) Work out: $\frac{1.2 \times 0.04}{0.6 \times 0.2}$

(03 marks)

b) Simplify: $\frac{4}{5} - \frac{1}{4} + \frac{1}{3}$ of 12

(03 marks)

28. In one day, a school prepares posho and beans in the ratio of 7:1 respectively. If 72 more kilogrammes of posho than beans are prepared.
a) How much posho is prepared for a day? **(03 marks)**

b) If the cost of each kilogramme of beans is Sh. 4000. How much will the school spend on buying the beans needed for a week? **(02 marks)**



29. Cathy saves thrice as much as what Betty saves per month. The difference in their saving is Sh.120,000.

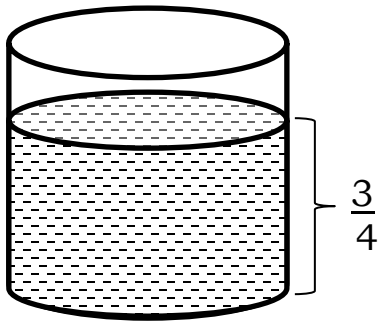
a)How much will Cathy save in one month?

(03 marks)

b)How much will Betty save in 4 months?

(02 marks)

30. The tank below can hold a full capacity of 200 litres. Some water was used to wash clothes.
Study it carefully and use it to answer the questions.



- a) How many litres of water were left in the tank after washing the clothes?
(02 marks)

- b) If 50 litres of water in the tank were also used to mop the house, what fraction of water finally remained in the tank?
(03 marks)



31. The table below represents the number of teams which participated in a tournament and the number of goals scored. Study it carefully and use it to answer the questions that follow:

Goals Scored	30	k	20	25
Number of Teams	2	1	3	4

a) How many teams participated in the tournament? **(01 mark)**

b) If the average number of goals scored was 26, find the value of k.
(03 marks)

c) How many teams scored below the average number of goals?
(01 mark)

32. Using a pair of compasses and a ruler only,
- a) construct a circle of diameter 8cm. **(01 mark)**
 - b) open your pair of compasses to a radius of the circle in (a) above and use that radius to make six arcs around the circumference of the circle. **(01 mark)**
 - c) Number the arcs from 1 to 6 and join arc 2 to 4, 4 to 6 and 6 to 2. **(01 marks)**
 - d) What special name is given to the shape formed after joining? **(01mark)**
 - e) Measure the length of one side of the shape formed. **(01mark)**

*****END*****

